Karim Zaghw

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Research Profile

I am an MSc student in Computational Neuroscience at the University of Tübingen. I study how learning agents can represent and use epistemic uncertainty, discover causal and hierarchical structure, and build models that support planning and generalization. My current research draws on reinforcement learning, Bayesian methods, and causal and hierarchical representation learning.

Education

MSc Computational Neuroscience 2024/10–Present
University of Tübingen Tübingen, Germany
Expected graduation: 03/2027.

BSc Computer Engineering 2016/09–2021/05
Cairo University Cairo, Egypt
GPA: 3.74/4.00.

Publications

Karim Zaghw, Peter Dayan, and Georgy Antonov (2026). *Weight a Moment: Risk-Aware Exploration with Bayes.* RLC 2026 / RLJ; paper.

Karim Zaghw, Andrew Pashea, Marc Pritsch, Wouter Nuijten, Karl Friston, and Lancelot Da Costa (2026). "Renormalising Generative Models for Active Inference: Foundations, Derivations, and Verification". Accepted, IWAI 2026; arXiv.

H. Osman**, Karim Zaghw**, M. Hazem, and S. Elsehely (2020). "An Efficient Language-Independent Multi-Font OCR for Arabic Script". In: *The International Conference of Digital Image Processing and Pattern Recognition (DPPR).* arXiv; presentation.

Research Experience

Student Research Assistant 2025/01–Present
Department of Computational Neuroscience, Max Planck Institute for Biological Cybernetics Tübingen, Germany

- Developed Bayesian Moment Learning, a model-free distributional reinforcement-learning method that represents epistemic uncertainty over risk-relevant moments and extends value-of-perfect-information exploration to moment-based risk objectives; work accepted at RLC 2026 / RLJ.
- Current project: developing Bayesian Monte Carlo tree search with Normal–Gamma edge posteriors, Bayesian Q-learning-style backups, and posterior-probability and value-of-computation stopping criteria.

Lab Rotations

Lab Rotation, Causal Abstraction for Overcomplete Representations 2026/07/01–2026/09/23
Robust Machine Learning Group, ELLIS Institute Tübingen Tübingen, Germany

- Studying when block-identifiable overcomplete representations preserve high-level causal abstractions under reparameterization.

Lab Rotation, Renormalising Generative Models 2026/04/15–2026/06/16
Methods of Machine Learning Group, University of Tübingen Tübingen, Germany

- Reconstructed and verified a hierarchical active-inference framework, translating core MATLAB/SPM routines into Python and clarifying its inference, planning, and structure-learning derivations; work accepted at IWAI 2026.

Teaching and Mentorship

Internship Mentor, Raisa Energy 2023/07–2023/10
Mentored a Text-to-SQL chatbot project using LLMs; achieved over 90% accuracy on daily queries; project write-up.

Junior Teaching Assistant, Operating Systems, Cairo University 2021/01–2021/06
Assisted in teaching, writing, and grading assignments, projects, and quizzes for 100+ students.

Additional Training

MITx MicroMasters Program in Statistics and Data Science 2023/02–2024/04
MITx (MIT Open Learning), edX; SDS General Track.

Deep Learning Specialization, DeepLearning.AI 2022/12–2023/05

Professional Experience

Data Scientist / Data Engineer, Raisa Energy 2021/07–2024/09
Built probabilistic forecasting, risk-modeling, and data-engineering systems used across 20+ projects and $1B+ of tracked assets.